

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

6 September 2026

Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Consider reaching for a cup. A visual description of its location and a description of the hand's relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identifi-

23 cation with perceptual content is a further modelling task.

24 The mathematical contribution is to specify connections between these
25 requirements and the assumptions each connection needs. A declared ac-
26 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
27 overlaps convert phase order into a bound on content disagreement. Compat-
28 ible spatial measures have a unique global extension; approximate agreement
29 controls a specified global selection. A contracting reconstruction makes that
30 extension its unique fixed point only if it accurately reconstructs the exten-
31 sion. Accuracy across a family of states imposes a further condition on the
32 information retained by the encoding.

33 The empirical contribution is to identify what observations can decide.
34 Stationary phase distributions can be shared by stronger coupling, weaker
35 diffusion and a matched common drive. Finite ramps change the observ-
36 able onset. Coherent adaptation can lose alignment with a designated en-
37 vironmental template, and a compatibility estimator needs diagnostics that
38 distinguish content from decoder artifacts. These findings determine the
39 measurements needed to test the proposed connections.

40 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
41 examines their observables, before Section 6 develops the cortical hypothesis
42 and an awakening experiment. The hypothesis is that extracellular contrac-
43 tion increases field-mediated coupling; its direction, rate and relation to re-
44 covery require independent calibration. Section 7 states the thermodynamic
45 composition and all eight connecting hypotheses per branch. Supporting

models of feedback, learning and resource supply have distinct state spaces
 and assumptions. Supplemental Material organizes their constructions and
 controls behind the claims they support.

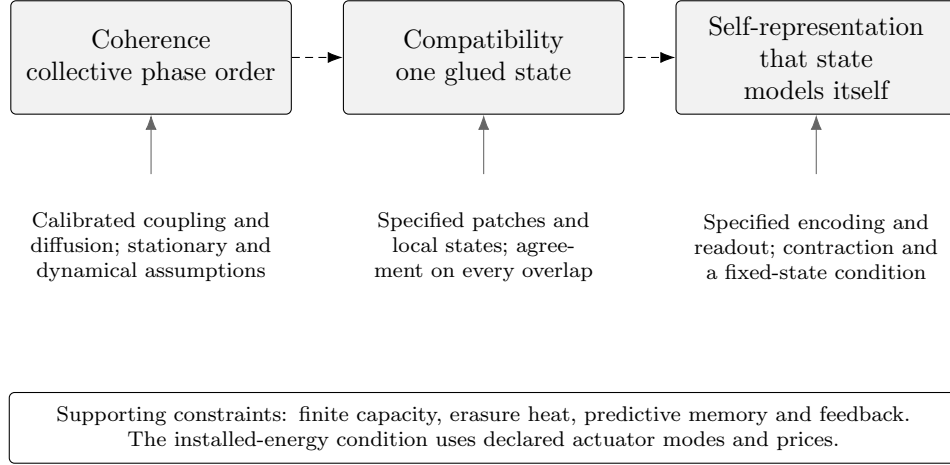


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

49 2 Coherence: physical coupling and its re- 50 source condition

51 2.1 Dynamics and the stationary threshold

52 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

53 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
54 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
55 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
56 parameters. The stationary calculation below uses identical frequencies and
57 this scalar coupling. Applying it to a spatial kernel requires an aggregation
58 rule and a check of the resulting approximation.

59 For identical frequencies in their rotating frame, the positive normalized
60 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

61 where I_n are modified Bessel functions and ψ is the collective phase. The for-
62 mal derivation uses the absence of stationary current for a periodic gradient
63 drift to obtain this density, then verifies normalization and self-consistency.
64 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one

65 positive solution coexists with zero and increases strictly with K . Its onset
 66 is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 Figure 2 shows this stationary branch.

68 Linearization about the uniform density gives rate $K/2 - D$ for the first
 69 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 70 exactly above threshold. About the coherent density, the quadratic form
 71 is strictly dissipative transverse to collective rotation, whose rate is zero.
 72 These linear estimates concern the limiting equation. Relating them to finite
 73 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 74 tion; exchange symmetry alone does not establish the needed independence
 75 (Sznitman, 1991).

76 2.2 Installed energy bounds mean coupling

77 A spatial model replaces the finite sum by integration against a kernel. One
 78 declared implementation carries response modes with continuous profiles ϕ_i ,
 79 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

80 The same configuration sets the reciprocal kernel and stored energy. An
 81 executed transition changes the kernel through its changed occupancies and
 82 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The

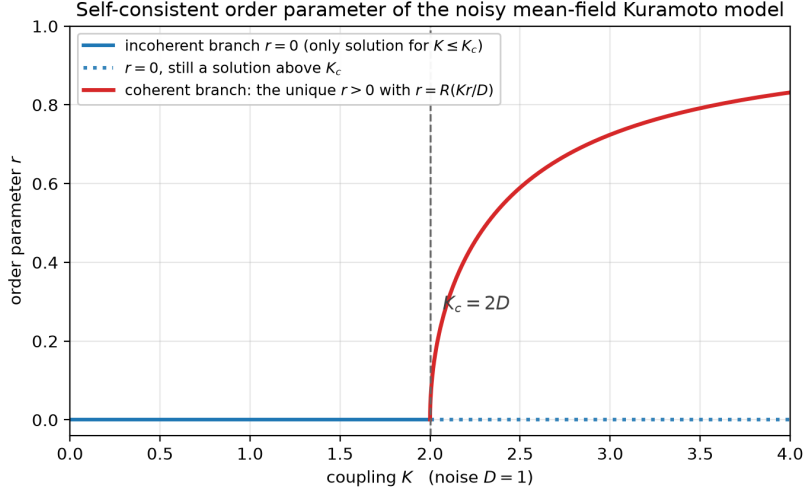


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

83 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 84 pair reconstruction on refining spatial partitions converges to its product-
 85 measure integral under the specified compactness and sampling assumptions;
 86 the refinement index is independent of the number of control updates.

87 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 88 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 89 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

90 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
91 coherent stationary branch and any growing Fourier mode about the uniform
92 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
93 is necessary: energy above it need not produce sufficient coupling because
94 the bound can be loose.

95 The quantity constrained is installed energy. Dissipated heat does not
96 determine it, and κ is supplied by the mode profiles, prices and substrate
97 normalization. Applying the scalar threshold to heterogeneous or signed spa-
98 tial dynamics requires a justified reduction. Identifying the installed kernel
99 with a cortical field is a separate physical commitment.

100 **3 Compatibility: from coordinated activity** 101 **to a global description**

102 **3.1 Why common timing is insufficient**

103 For overlapping patches U and V , compatibility means that restricting either
104 local description to $U \cap V$ gives the same object. Descriptions can agree on
105 shared quantities while contributing different information elsewhere. Com-
106 patible sections of the specified sheaf have a unique global extension. On
107 a finite cover, local finite spatial measures agreeing as measures on every
108 overlap also have a unique finite-measure extension whose sheaf image is the
109 glued section. No mass-one normalization is imposed.

110 In the cup example, visual and reaching descriptions must express a
 111 shared quantity, such as cup location, in common coordinates. Restriction
 112 retains that quantity while dropping details particular to each description.
 113 Agreement permits surface appearance and hand configuration to contribute
 114 to one situation. The mathematical counterexample assigns the same phase
 115 to every patch while two local mass profiles disagree on a shared site. Those
 116 profiles have no common extension. Because they are chosen independently
 117 of phase, synchrony places no constraint on their disagreement.

118 This distinction limits the biological interpretation. Receptive territories
 119 and content domains must be related to spatial patches, and the descriptions
 120 must support the required extension. Matching overlap marginals of de-
 121 coded probability laws need not determine a unique global joint distribution
 122 (Abramsky & Brandenburger, 2011). The spatial-measure result supplies a
 123 precise construction; applying it to perceptual content requires a justified
 124 content model.

125 **3.2 When coherence does constrain content**

126 A declared relation between phase and content gives a positive result. Let
 127 contents take values in a real normed vector space E . Patch i reads an
 128 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 129 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 130 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 131 sites; their equality on overlaps is a supplied relation between descriptions of

132 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 133 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 134 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

135 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 136 agree exactly. The factor N makes this estimate potentially weak in large
 137 populations; it supplies a pointwise guarantee from a population average. A
 138 shared scalar encoder is a special case. Dropping the overlap relation permits
 139 disagreement even at perfect locking. A common nonexpansive linear projec-
 140 tion preserves agreement, but agreement of a projection does not establish
 141 agreement of the full vector.

142 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 143 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 144 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 145 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

146 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 147 duce an initial mismatch. Approximate observations leave a nondecaying
 148 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 149 update to the phase process. These are mechanisms for specified vector con-
 150 tents in shared coordinates; their neural meaning and relevance to experience

151 remain to be established.

152 **3.3 What approximate agreement determines**

153 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
154 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
155 Weights that vanish outside their patches and sum to one at each site select
156 a global profile by weighted averaging. The selection lies within ε of each
157 patch there, and any two such selections differ by at most ε , independently
158 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
159 $L\varepsilon$ across selections.

160 At zero tolerance, the selection is the unique exact extension. At every
161 positive tolerance there are families with no exact extension; selection then
162 gives an approximation whose dependence on weights is bounded. Supple-
163 mental Material gives a three-site witness attaining the bound. The mass-
164 profile construction uses the scalar case of the content dynamics and addi-
165 tionally requires nonnegativity of the initial profiles and observations. The
166 general vector result supplies no identification with the spatial-measure sheaf.
167 Neither construction selects the neural cover or decoder.

168 4 Self-representation: a state that models it- 169 self

170 4.1 The reflexive construction

171 Compatibility gives a global state. Reflexivity relates it to an encoding within
172 that state. Equip global sections with a complete metric. If the additional
173 restriction-resonance condition identifies encoding with restriction to a region
174 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

175 The region and readout are fixed across the state space. If F contracts and
176 fixes the section s_* formed from the local descriptions, Banach's theorem
177 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
178 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
179 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
180 one for $\tau > 0$. Constructing a map with that factor remains a modelling
181 obligation.

182 In the cup example, the global state includes the represented scene and the
183 perceiver's relation to it. A local encoding must retain enough information
184 for the readout to recover that modelled state; cup position alone would be
185 insufficient if hand configuration and viewpoint could not be reconstructed.
186 Here *prediction* names a reconstruction map. Interpreting it as temporal

187 neural prediction requires dynamics.

188 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
189 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

190 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
191 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
192 weakens. A biological interpretation must specify the represented variables,
193 metric and acceptable fidelity; changing readouts need a further model.

194 4.2 Reconstruction across a family of states

195 Sensitivity can be required quantitatively. Declare relevant macrostates, an
196 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
197 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

198 Thus M relevant states pairwise more than 2ε apart require at least M
199 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
200 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
201 existing self-prediction map.

202 This is an additional accuracy criterion on a family, with no requirement
203 for multiple exact fixed points. A blind encoder or constant readout fails it
204 for separated states despite having a unique fixed point. The family, metric

205 and tolerance are inputs; interpreting the code count as physical memory
206 capacity requires a hardware model.

207 **4.3 Agreement after a change takes communication**

208 Snapshot compatibility does not specify how quickly descriptions respond
209 when a shared quantity changes. In a synchronous network whose updates
210 read only the site’s own state and declared neighbours’ messages, two initial
211 configurations agreeing on a site’s causal past through T rounds produce the
212 same state there at round T . If two fresh interventions outside that past
213 require reports separated by more than 2ε , no report from that site can be
214 within ε of both. This follows from the reconstruction bound in Section 4.2.

215 A three-site delay line witnesses failure after one round and exact report-
216 ing after two; cutting the path prevents delivery at every round. A report
217 may still happen to be correct about one fixed world. The bound concerns
218 responsiveness across interventions. Applying it requires a justified commu-
219 nication graph, observation paths and deadline.

220 **5 What the observations establish**

221 The three requirements concern different objects: phase dynamics, local de-
222 scriptions and internal encodings. Their observables must retain these dis-
223 tinctions. The following controls show which mechanisms remain indistin-
224 guishable, which temporal approximations fail, and what independent con-

225 tent measurements must establish.

226 **5.1 Stationary distributions leave the mechanism un-** 227 **determined**

228 Equation (2) gives a concentration–coherence relation with no fitted shape
229 parameter. Estimate r from circular resultant length and a from density
230 shape: inverting R from the measured r would make agreement automatic.
231 Even independently estimated agreement identifies a statistical family shared
232 by different mechanisms.

233 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
234 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
235 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
236 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
237 and diffusion D_0 have that same density. Their stationary distributions co-
238 incide window by window despite different causes.

239 Resolved dynamics supply additional information. In the ideal model with
240 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
241 increment means constrain drift. Separating endogenous coupling from a
242 matched common drive additionally requires a measured drive or controlled
243 perturbation; identifying electromagnetic coupling requires evidence specific
244 to that interaction. Supplemental Material gives the equivalence construction
245 and its numerical check.

246 5.2 Finite ramps change the observable onset

247 The square-root law in Eq. (3) describes stationary solutions. A changing
248 coupling requires time for the phase distribution to approach those solutions.
249 In a deterministic mean-field control that crosses the threshold symmetrically,
250 none of the scanned rates meets the declared tracking tolerance: slowing
251 the crossing does not remove the terminal residual under that initialization.
252 A slow control variable therefore supplies no general guarantee of branch
253 tracking at onset. Supplemental Material gives the residual, rate range and
254 admissible rates away from threshold.

255 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
256 at four coupling speeds. Escape is the first of three consecutive saved samples
257 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
258 replicas escape within the fastest window. The three slower ramps have
259 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
260 differ from those of the exact stationary branch evaluated on the same grids.
261 At the fastest speed the ensemble mean never reaches the fit window, so its
262 fitted slope does not estimate stationary onset.

263 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
264 makes the apparent exponent depend on ramp speed and fitting window.
265 The specified deviation-based critical ramp speed is not bracketed by this
266 sweep. With fluctuating coupling, the relevant stationary comparison also
267 depends on rate: the mean coupling approximates order in the tested fast
268 regime, while averaging stationary order over coupling works for the tested

269 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 270 neither approximation. These are finite-horizon observations, with controls
 271 and parameters in Supplemental Material.

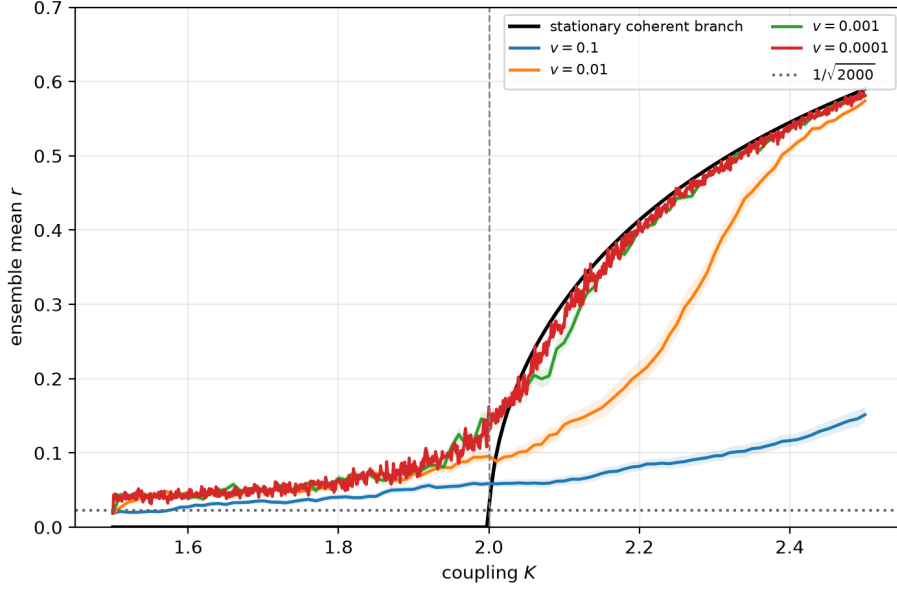


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

272 Additional controls address integration bias, population size and spatial
 273 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 274 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
 275 rameters and controls in Supplemental Material; they supply no cortical cal-
 276 ibration.

277 **5.3 Coherent adaptation can lose environmental align-** 278 **ment**

279 An adaptive coupling rule might supply representational structure, but its
280 objective must reward the relevant agreement. For symmetric coupling, pair-
281 wise sine terms cancel in the population mean drift. Minimizing squared drift
282 at that fixed mean therefore rewards equalizing drift across nodes. In the
283 frequency-cluster control, this favors coupling between unlike-frequency clus-
284 ters, whereas the diagnostic environmental template rewards coupling within
285 clusters.

286 The formalization proves descent of the squared-drift objective for a spec-
287 ified gradient flow at fixed phases. Its functional differs from thermodynamic
288 entropy production: at a coherent stationary density the probability current
289 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
290 rial gives the operator, its gradient and the conditions of the descent result.

291 The finite control co-evolves noisy phases and symmetric, nonnegative
292 couplings at fixed total resource. Its template is a diagnostic covariance of
293 latent signals, never a training input. Across three seeds, second-half order
294 remains between 0.9167 and 0.9222, while the within-cluster to between-
295 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
296 Random-update, relabelled-gradient and frozen-kernel controls are reported
297 in Supplemental Material.

298 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

5.4 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built

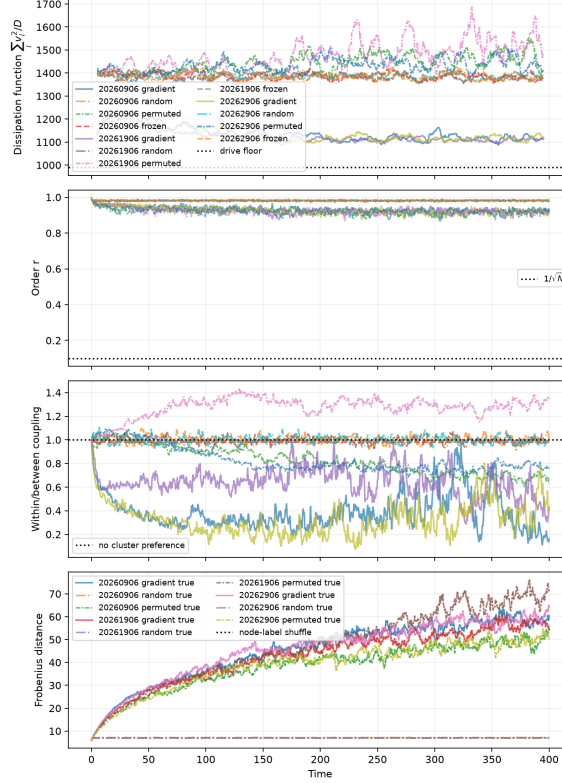


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

321 from it and 0.000 when it is not. A phase-derived statistic reports com-
322 patibility on phase-locked patches whose local descriptions disagree. The
323 implemented content statistic detects their incompatibility. These controls
324 validate a measurement on known constructions; a neural content model and
325 decoder remain to be established.

326 Choosing the cover also chooses a candidate subject: trivial descriptions
327 can agree while carrying little content, and a partition can conceal disagree-
328 ment. The useful choice must therefore be constrained by what neural ac-
329 tivity encodes. The exact composition assumes compatibility of its supplied
330 cover. The content mechanisms above identify additional hypotheses under
331 which agreement can arise; they leave the empirical choice of content and
332 the connection to that cover explicit.

333 5.5 Phase-density estimation and measurement feasi- 334 bility

335 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
336 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
337 itages produce different residuals. The bipolar results are compatible with the
338 Bessel relation, but their concentration range cannot distinguish it from its
339 linear approximation at the observed scatter. Temporal pooling changes the
340 estimand, and sedation recordings leave the awakening hypothesis untested.

341 The concentration range and dependence determine whether the relation

can be tested. Under the specified log-density estimator and bias-floor criterion, discrimination at the observed concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned designs, rising to 31000 when sites share a mean in groups of 10. These counts describe that estimator and dependence model; the intended recording requires its own calibration. Supplemental Material reports the bin-count controls and the full dependence-aware design.

An informative dataset needs sufficient concentration range or precision, calibrated estimators and an observation model for dependence, source mixing and reference choice. Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

6 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 5 determine what observations must distinguish before a phase trace can support this identification.

361 6.1 Physical identification and calibration

362 The cortical hypothesis identifies part of effective coupling with endogenous
363 extracellular electromagnetic interactions. Fields can modulate spike timing
364 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
365 propose coordination of distributed neural information (Pinotsis & Miller,
366 2022; McFadden, 2020). Phase organization under propofol motivates study-
367 ing this regime (Bastos et al., 2021), but does not measure coupling.

368 The formal field predicate requires a jointly continuous kernel, a normal-
369 ized substrate, coupling equal to the kernel average, and no single site pro-
370 viding more than half that strength. Toy constructions satisfy it. A cortical
371 realization must specify the scale at which the kernel and its reduction ap-
372 ply: calcium imaging reports different state-dependent modularity patterns
373 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

374 The spatial reduction is also testable. A row-sum aggregation at zero
375 diagonal tracks the uniform-coupling control over the tested spatial decay
376 range, while frequency heterogeneity displaces the operational threshold. The
377 fit window biases both estimates relative to $2D$; agreement with the control
378 is therefore the supported conclusion. Supplemental Material reports the
379 reduction, its numerical error and the distinct assumptions needed for the
380 kernel-average identification in Eq. (5).

381 The field commitment requires physical evidence beyond a choice of math-
382 ematical representation. A finite matrix spread over positive-mass cells pre-
383 serves its total coupling and phase correlation exactly. Point-supported

384 vanishing under an atomless measure and a disadvantage from excluding
 385 a better-correlated pair concern specified support constraints. Neither com-
 386 parison supplies energy prices or a biological–digital distinction.

387 A spike-time shift alone supplies no coupling rate. The dimensionless
 388 product of population size N , timing sensitivity s , field amplitude E and
 389 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (11)$$

390 This rate describes how induced timing shifts accumulate into drift. Nei-
 391 ther it nor the installation factor κ follows from a heat bound. Testing the
 392 physical threshold requires independent estimates of the interaction kernel,
 393 its aggregation and diffusion in the same cortical regime.

394 **6.2 Extracellular geometry and recovery**

395 The proposed control variable is extracellular geometry. Extracellular vol-
 396 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 397 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 398 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 399 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 400 an effective-medium description predicts that a larger extracellular fraction
 401 raises conductivity and reduces field amplitude.

402 The coupling-gated hypothesis consequently predicts that extracellular

403 contraction on awakening raises E and, holding the other factors in Eq. (11)
 404 fixed, raises effective coupling. This argument supplies a direction; the
 405 geometry–coupling relation, its magnitude and its role in human recovery
 406 require measurement.

407 For a smooth crossing at nonzero speed, evaluating the stationary branch
 408 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (12)$$

409 This follows from Eq. (3) at fixed D and concerns the stationary branch
 410 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 411 dently justified tracking approximation. The crossing controls in Section 5.2
 412 leave that approximation unsupported at onset, even though a separation of
 413 geometry and phase time scales can support it away from threshold. The
 414 awakening hypothesis therefore needs a calibrated dynamical prediction for
 415 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 416 does not supply a recovery latency.

417 **6.3 A protocol for the awakening experiment**

418 The physical test has six requirements:

- 419 1. **Coupling and diffusion in the same units.** Estimate them from re-
 420 solved phase dynamics, separating heterogeneity and observation noise.
 421 Measure the drive or perturb the system to distinguish endogenous cou-

422 pling.

423 **2. Geometry calibrated to coupling.** Validate the geometry observ-
424 able and its relation to effective coupling, with uncertainty. The con-
425 ductivity argument supplies only the expected direction.

426 **3. Calibrated phase measurements.** Estimate r by resultant length
427 and a by unweighted regression of log density on centered cosine.
428 Choose site count and binning from a dependence-aware calibration
429 over the intended concentration range. The tested designs require
430 thousands of sites at the observed concentrations; temporal pooling
431 changes the spatial estimand.

432 **4. An independent recovery endpoint.** Measure behavioral recovery
433 separately from phase organization.

434 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
435 old above the finite-size floor and evaluate its latency under the mea-
436 sured coupling trajectory, including ramp speed or fluctuations.

437 **6. Matched model comparison.** Compare coupling increase, diffusion
438 change, common drive and alternative recovery models under the same
439 observation and filtering assumptions.

440 Failure of the concentration–coherence relation over an informative range
441 constrains the density model. Failure of the calibrated geometry–coupling
442 association constrains the physical gate. Better prediction of independent

recovery by a competing mechanism counts against its proposed role. A favorable outcome still leaves the content of coordinated activity to be explained.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Laudauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still et al., 2012). Zero retained information also satisfies it, so a task and a learning rule require additional specification. Relating this bound to Lan-

461 dauer heat requires a common thermal scale and budget, and memory waste
 462 bounded by erased entropy.

463 An acting agent changes the signal it will receive. Specify policy, world,
 464 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{14}$$

465 The second conditional term vanishes in the passive model. With feedback,
 466 the signed difference can be negative as action creates correlations with the
 467 future. A causal reading requires an environmental state sufficient to describe
 468 the relevant dynamics.

469 For a controlled substep, the active thermodynamic bound instead uses
 470 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 471 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 472 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 473 2013). Every forward path of positive probability must have a positive-
 474 probability reverse for this finite identity. A missing reverse gives infinite
 475 divergence. The supplement develops the support conditions and composes
 476 actuation with memory updates on their shared intermediate law.

477 **7.2 Learning and repeated operation need explicit re-** 478 **sources**

479 The finite examples distinguish selecting a feasible policy, relaxing a register
480 toward an encoded reward, and learning an unknown task from observa-
481 tions. The observational learner acquires correlation with an environmental
482 parameter while its action marginal stays uniform. A continuing model exe-
483 cutes actuation, observation-driven learning and reset in sequence, preserving
484 above-chance performance at a positive work cost per cycle.

485 A phase-derived sensor makes the informational distinction concrete: an
486 informative readout and a blind control have the same phase order but dif-
487 ferent learning performance. Attaching the learner’s register to the actuator
488 of Section 2.2 ties reward, installed kernel and work to its own executed law.
489 The learning witness rises above the necessary installed-energy bound while
490 its matched control remains on the exclusion line. These finite constructions,
491 detailed in Supplemental Material, leave the evolving feedback between phase
492 dynamics, installation and sensing to be specified.

493 For that model, common energy E , thermal scale θ , local detailed balance
494 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (15)$$

495 where expected cumulative work stays within the supplied allowance B at
496 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

Table 1 (continued)

Connection	Status	Required assumption
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

515 Both branches also take a thermodynamic cover carrying exact over-
516 lap compatibility and a reflexive boundary specifying encoding and readout.
517 They conclude **UnifiedSelf**: the glued section is the unique fixed point of
518 self-prediction. Compatibility and accuracy at that section enter as premises;
519 uniqueness follows. Identifying encoding with restriction needs the separate
520 resonance condition of Eq. (8).

521 Joint witnesses combine a register, boundary, refining mesh and small
522 spatial cover; active witnesses add controlled thermal updates and a priced
523 installation. They establish satisfiability under their stated identifications.
524 Mesh convergence follows from refinement data, and the witnesses establish
525 no common cortical mechanism. Supplemental Material maps the claims to
526 their Lean declarations and records the counterexamples. The development
527 declares no additional axioms beyond its libraries’ foundational axioms.

528 8 Discussion

529 8.1 Unity and a minimal self as empirical commit- 530 ments

531 We propose the state glued from compatible local descriptions as a corre-
532 late of conscious unity, and its accurate internal reconstruction as a correlate
533 of a minimal self. The motivations are that different contents belong to
534 one experienced situation and that the situation is experienced from a per-
535 spective. The conditional results identify relations a realization would need:
536 sufficient installed resources for the specified phase model, agreement about
537 shared content, and an encoding that supports reconstruction at the required
538 fidelity.

539 Compatibility allows different descriptions to contribute different infor-
540 mation. It also permits uncertainty and conflicting judgments: apparent cup
541 position and judged cup position can be distinct quantities in a consistent
542 description of their discrepancy. Which quantities are shared must be de-
543 termined independently. Choosing them solely to secure agreement would
544 deprive the proposal of empirical force.

545 The minimal-self proposal concerns the perceiver’s position and condition
546 within the situation, such as the cup’s location relative to a body that can
547 reach it. The encoding could be implicit and need carry no autobiograph-
548 ical narrative. Its proposed necessity is philosophical: even an unreflective
549 episode has a perspective. Gluing by itself requires no self-encoding, and re-

constructability alone supplies no functional role. Altering the encoded hand or viewpoint while holding the external scene fixed should change estimates of the perceiver’s relation to the cup as the readout predicts. Whether that represented perspective is experienced remains a further question.

8.2 What a test of the full proposal must establish

Both uniqueness results concern specified objects. They choose neither the cover nor the encoding, and do not select between candidate subjects. Integrated information theory treats exclusion explicitly (Tononi et al., 2016); here selection requires independent evidence about neural organization. A constant readout can contract and fix a predetermined state while discarding its input. The family criterion in Section 4.2 tests the sensitivity that this construction lacks, provided the relevant states and fidelity are declared independently.

A test must separately measure coupling relative to diffusion, content compatibility and self-reconstruction, alongside an independent endpoint for experience. Meeting all three without a unified episode would challenge sufficiency. A unified episode with incompatible descriptions would challenge the unity identification; one lacking the specified self-encoding would challenge the reflexive requirement. Awakening, dreaming and dissociative states offer possible settings, but absence of report cannot by itself define absence of experience.

The next physical requirement is a calibrated geometry–coupling relation

572 and resolved phase dynamics. The installation bound contributes a nec-
573 essary resource condition where its hardware representation applies; cortical
574 mode prices and field-mediated coupling require measurement. For the repre-
575 sentational connection, an informative experiment asks whether a controlled
576 change in coordination alters independently decoded content agreement, with
577 controls for sensory information and decoder quality. The constructed-data
578 estimator establishes feasibility for specified contents. Neural local descrip-
579 tions, shared overlap encoders and the self-encoding have yet to be identified.
580 The population-size dependence of the phase-to-content bound further limits
581 its quantitative use.

582 The separation results explain why these measurements must be inde-
583 pendent. A stationary distribution leaves several recovery mechanisms indis-
584 tinguishable. Equal phase order can accompany incompatible descriptions
585 or different task information. The tested plasticity rule preserves coherence
586 while losing its designated template. Each finding concerns the specified
587 model and observable; their role is to delimit the evidence needed for the
588 proposed connections.

589 **8.3 Explanatory scope**

590 Integrated-information, global-workspace and predictive-processing accounts
591 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
592 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
593 candidate physical interaction and conditional relations between local content

594 and self-encoding. Gluing and reconstruction permit multiple substrates;
595 the electromagnetic commitment enters through physical calibration. The
596 formalization checks implications under their hypotheses, and its witnesses
597 establish realizability on declared models. They do not together supply a
598 cortical agent.

599 The explanatory question is whether the specified relations occur in neural
600 activity and matter for experience. In the cup example, this means establish-
601 ing how coordination helps maintain one represented situation and how the
602 perceiver’s place within it contributes to the episode. The proposal identi-
603 fies candidate correlates and tests of that identification. Why those relations
604 would be accompanied by experience remains unexplained.

605 **Data and code availability**

606 The Lean development, simulation scripts, tests and saved numerical sum-
607 maries are maintained in the manuscript source repository. Publication
608 simulation values are generated from saved summaries without rerunning
609 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
610 ds005620 (Bajwa et al., 2025); preprocessing and selection are described in
611 Supplemental Material. The repository contains the reproducible analysis
612 and does not redistribute the raw recordings.

613 **Funding and competing interests**

614 This research received no external funding. The author declares no compet-
615 ing interests.

616 **Use of AI tools**

617 AI assistance to this work was substantial and is not separable by task. Sev-
618 eral coding assistants and chat interfaces were used in an interleaved way
619 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
620 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
621 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
622 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
623 accessed with the Hermes CLI. Between them they contributed to the Lean
624 development, the simulation and analysis code, and the drafting and revision
625 of both documents. Because the tools were used in combination and in alter-
626 nation, no section, proof, figure or result is attributable to any one of them,
627 and none is claimed to be. The author is responsible for all content, including
628 everything a tool produced. Machine checking establishes the stated Lean
629 results under their hypotheses; it does not validate biological interpretations
630 or replace scientific review of generated code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How particular is the physics of the free energy principle?. *Physics of Life Reviews*, 40, 24-50.
- Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some parts of the free energy principle. *Entropy*, 23(3), 293.

650 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
651 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

652 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
653 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

654 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
655 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
656 *Progress in Neurobiology*, 226, 102465.

657 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
658 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
659 9(8), 45-60.

660 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
661 testable hypothesis about the characteristics of conscious as opposed to
662 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

663 McFadden, J. (2020). Integrating information in the brain’s EM field: the
664 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
665 niaa016.

666 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
667 guide neocortical network activity. *Neuron*, 67(1), 129-143.

668 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
669 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
670 95-103.

- 671 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
672 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 673 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
674 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 675 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
676 *logical Reviews*, 98(1), 239-389.
- 677 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
678 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
679 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 680 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
681 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
682 ity of complexity measures to capture dreaming during propofol sedation.
683 *Scientific Reports*, 15, 32746.
- 684 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
685 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
686 brain. *Science*, 342(6156), 373-377.
- 687 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
688 *Physiological Reviews*, 88(4), 1277-1340.
- 689 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
690 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

- 691 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
692 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
693 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 694 Landauer, R. (1961). Irreversibility and heat generation in the computing
695 process. *IBM Journal of Research and Development*, 5(3), 183-191.
- 696 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
697 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 698 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
699 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 700 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
701 of prediction. *Physical Review Letters*, 109(12), 120604.
- 702 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
703 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- 704 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
705 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
706 031015.
- 707 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
708 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
709 PhysRevLett.111.180603.

- 710 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
711 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 712 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
713 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
714 181.
- 715 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
716 archive.pitt.edu/115.
- 717 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
718 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
719 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 720 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
721 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
722 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 723 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
724 bifurcations with additive noise. *Probability Theory and Related Fields*,
725 122, 341–388. doi:10.1007/s004400100174.
- 726 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
727 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
728 doi:10.1088/1367-2630/13/11/113036.
- 729 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
730 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

731 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
732 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

733 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
734 terns underlie complex brain dynamics and are related to cognitive pro-
735 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
736 023-01626-5.

737 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
738 and dissipation with partially observed fluctuating signals. *Physical Review*
739 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

740 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
741 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
742 cell resolution functional networks during unconsciousness are segre-
743 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
744 doi:10.1016/j.celrep.2025.116902.

745 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
746 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
747 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.